

Optimization:

↳ Optimization in machine learning is the process of adjusting a model's internal parameter to minimize or maximize an objective function.

Optimization Process:

1. Forward Pass:

The model takes input data and generate a prediction.

2. Loss Computation

A loss measures the diff. b/w prediction and true answer/value.

3. Backward Pass: (Optimization)

The optimizer calculates the derivative (gradient) of the loss with respect to each parameter and adjusts them to reduce the error in next iteration.

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \cdot \nabla L(\theta)$$

θ : model parameter

η : learning rate

∇L : Gradient of loss function

Type of optimization algo.

1) Gradient Based optimizers (First order)

Optimizer	How it work	Use
(a) Batch Gradient Descent (BGD)	Calculate gradient using entire dataset per step	Small Dataset
(b) Stochastic Gradient Descent (SGD)	Update weight after every single sample. (Faster but noisy)	Large Dataset helps escape local minima
(c) Mini-Batch SGD	Updates weights using small random subset	Train ANN

2) Adaptive Learning Rate Methods:

- ↳ Standard gradient method use a fix step size
- ↳ while adaptive optimizers dynamically change learning rate for each specific parameter based on past steps.

↳ used in NLP.
Optimize

(a) RMSprop

How it work

Use.

Adapts the learning rate individually for each parameter based on historical updates

NLP

(b) Adam

Combine momentum & RMSprop. It combines exponentially average of both gradient and square gradient.

DL &
LLM

3) Second Order optimization

↳ These algorithm use both first derivative (gradient) and second derivatives (Hessian matrix)

↳ It converges faster than first order method

↳ Computationally more expensive

a) Newton method

↳ find local minimum or maximum of a function

↳ This method use first and second order derivatives to update guesses for optimum

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

Steps:

1. Initialization : start with initial guess (x_0)

2. Gradient & Hessian : calculate gradient $f'(x)$ & Hessian $f''(x)$

3. Update : use iteration formula to compute next approximation.

4. Convergence check: Repeat until convergence

↳ Fast quadratic convergence near ~~no~~ minimum
no need to manually tune a learning rate

Con: ① time take $O(N^3)$ for $N \times N$ Hessian matrix
② impractical for deep learning or large dataset

(b) Constrain optimization

↳ optimize objective function with certain restrictions or constraints.

↳ These constraints can equality or inequality conditions.

↳ Specific method help find optimal solns

Ex Lagrange multipliers introduce additional variable called multiplier to convert a constrained problem with equality constraints into an unconstrained one.

(c) Bayesian optimization

↳ It is probabilistic method used to optimize expensive or complex functions.

↳ It uses past evaluation to decide where to search next.

Optimization algo in ML

First-Order Algos

- Gradient Descent
- Stochastic optimization
- Evolutionary algo.
- Metaheuristic optimization
- Swarm Intelligence algo.
- Hyperparameter optimization
- Optimization in deep learning

Second Order Algo.

- Newton method
- Quasi " "
- Constrained optimization
- Bayesian " "

Evolutionary Algo:

↳ It is inspired by natural selection

↳ They work by evaluating a group of candidate of solⁿ over multiple generation to find good or near optimal solⁿ for complex problem.

↳ Main component of Evolutionary Algos

- ~~the~~ Population: A group of possible solⁿ
- Fitness Function: measures how good each solⁿ is
- Selection: Chooses the better solⁿ for reproduction
- Genetic operation: create new solⁿ using crossover and mutation
- Terminate: stop process after condition met.

Ex Genetic Algorithm (GA) and Differential Evolution (DE)